Fiscal Year:	FY 2011	**Task Last Updated:**	FY 01/25/2022
PI Name:	Menon, Naresh Ph.D.		
Project Title:	Novel Fluid Preservation System		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	(626) 381-9974
Organization Name:	ChromoLogic LLC		
PI Address 1:	1225 S. Shamrock Ave		
PI Address 2:			
PI Web Page:			
City:	Monrovia	**State:**	CA
Zip Code:	91107-4244	**Congressional District:**	32
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	06/02/2011	**End Date:**	06/01/2013
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX11CB42C		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	To address NASA's need for a method to preserve human biological samples, mostly blood and urine, from astronauts collected during flight under ambient conditions, ChromoLogic (CL) has developed a novel Fluid Preservation System (FPS). FPS is based on collecting and sealing fluids in a sterile, hermetically sealed volume, with automatic separation of supernatant fluid where necessary. The unique microfluidic and medical expertise of CL scientists has resulted in an innovative and lightweight fluid storage system that utilizes compact and rugged microfluidic chips and novel valve technology that are capable of processing ~10 ml of samples in a few minutes using on-board processing equipment. In Phase I CL has demonstrated the feasibility of the FPS technology by building prototype chips and demonstrating the proof of concept of pumping, separation, storage, and preservation. The FPS has been demonstrated to be able to store blood plasma at room temperature for 54+ days. In Phase II CL will develop a fully functional system that will be FDA (Food & Drug Administration) approved and space qualified.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential Non-NASA commercial applications exist in fields such as blood sample collection and storage by medical personnel operating in locations far from traditional preservation infrastructure. From third world counties to rural America medical practitioners that serve outlying areas are often constrained by the lack of a method of blood preservation; reducing the need to carry methods of refrigeration will increase the total amount of samples that can be collected and will increase the efficiency of labor force used for such work. Other fields that will be targeted by the Fluid Preservation System (FPS) are research and development environments. Civilian research applications include large epidemiological studies and remote field research, such as environmental research. There also exists many opportunities for use of the FPS by the military for bio-sample collection and management under challenging field conditions.
Task Progress:	New project for FY2011. Reporting not required for this SBIR Phase II project. (Ed. note: added to Task Book January 2022)
Bibliography Type:	Description: (Last Updated:)